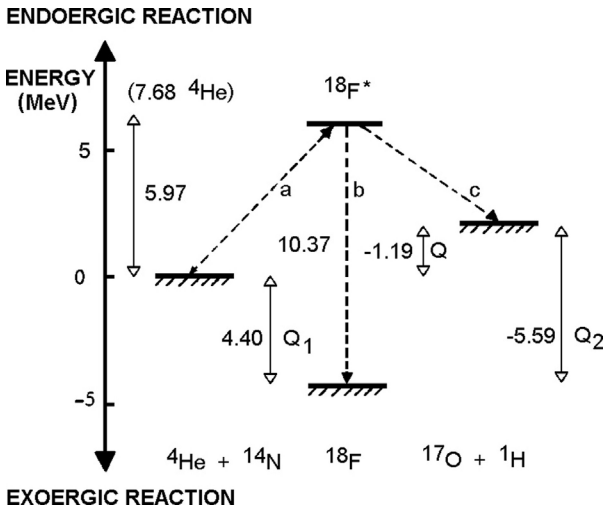


Figure 10.7 Energy diagram of the transmutation $^{4}He + {}^{14}N \rightarrow {}^{18}F^{*} \rightarrow {}^{17}O + {}^{1}H$ caused by bombarding ^{14}N with 7.68 MeV α-particles from ^{214}Po.

If the projectile has a higher kinetic energy than the threshold energy, the products would have a correspondingly higher combined kinetic energy, assuming no transformation into other forms of energy. Rutherford used α-particles from the radioactive decay of ^{214}Po, which have kinetic energies of 7.68 MeV. The products, therefore, had considerable kinetic energy, which is shown by the thick tracks in the cloud chamber picture of Figure 10.6.

In Figure 10.7 the energy pattern for our reaction is summarized. The Q-value for the first step of the reaction ($^{4}He + {}^{14}N \rightarrow {}^{18}F$) is $+4.40$ MeV (Q_1). The kinetic energy of the ^{4}He is 7.68 MeV, so conservation of momentum requires that the ^{18}F nucleus have $7.68 \times 4/(4 + 14) = 1.71$ MeV of kinetic energy. Thus, the internal excitation energy of $[{}^{18}F]^{*}$ is $4.40 + 7.68 - 1.71 = 10.37$ MeV. For the second step, $^{18}F \rightarrow {}^{17}O + {}^{1}H$, the Q_2-value is -5.59 MeV. This shows that the total excitation and kinetic energy of the products is $10.37 + 1.71 - 5.59 = 6.49$ MeV.

10.8. THE COMPOUND NUCLEUS MODEL

We have used a model which assumes the formation of an excited intermediary nucleus in the previous section to discuss some aspects of nuclear reactions. Let us now study this model a little closer.

If $E^{0}_{\text{kin}} > E_{\text{cb}}$ the attractive nuclear force dominates and the particle is absorbed by the target nucleus. Assuming $Q > 0$, the E^{0}_{mass} decreases. This means that E_{exc} increases, and the system is transformed into an excited compound nucleus. Moreover, since the projectile had a certain kinetic energy, the excitation of the compound nucleus is

$$E_{\text{exc}} = Q + E^{0}_{\text{kin}} \tag{10.29}$$

This equation is based on the assumption that the compound nucleus is the center of reference. It should be noted that the height of the Coulomb barrier does not influence the excitation energy of the nucleus in any way other than that a projectile must have a kinetic energy greater than that value before the reaction can occur. In the reaction all E_{kin} and E_{coul} are transferred into E_{kin}, E_{exc} and E^0_{mass} for the compound nucleus. The excitation of the compound nucleus can be removed either through the emission of γ-rays, neutrons, or by the decay of the compound nucleus into different product nuclei. In the former case, $E_{kin} = E_\upsilon$. In the latter case we again have to bring in (10.10).

If the reaction passes through formation of a compound nucleus, having a lifetime of many nuclear vibrations, we should expect that the amounts and types of reaction products were independent of the projectile − target combination used to produce a specific compound nucleus and only depend on its excitation energy. Essentially the compound nuclear does not have any memory of how it was formed and its decay is controlled by its own features only. This has indeed been found to be the case for some kinds of nuclear reactions, see Ch. 11.

10.9. RADIOACTIVE NEUTRON SOURCES

Neutrons were discovered through the reaction between α-particles, emitted by radioactive substances, and light elements: ^{4}He (from Po) + ^{9}Be → ^{12}C + n (Chadwick 1932). The coulomb barrier of light elements is sufficiently small to be penetrated by α-particles emitted by radioactive nuclides; in the reaction used by Chadwick the coulomb barrier is only ∼3.5 MeV, eqn. (10.15) and Fig. 10.3. All early neutron research was conducted with sources of this kind, the most popular being the RaBe mixture. A number of radioactive n-sources are listed in Table 10.2.

Table 10.2 Radioactive neutron sources

Material	Half-life	Neutron yield (n s^{-1} Bq^{-1})	γ-dose rate[(a)] (mSv h^{-1} m^2 Bq^{-1})
^{226}Ra + Be	1600 y	3.5×10^{-4}	850
^{239}Pu + Be	24110 y	2.4×10^{-4}	4
^{239}Pu + ^{18}O	24110 y	7.8×10^{-6}	[(b)]
^{241}Am + Be	433 y	6.8×10^{-5}	2.5
^{210}Po + Be	138 d	6.8×10^{-5}	<0.3[(c)]
^{124}Sb + Be	60 d	$\sim5\times10^{-6}$	1000
^{252}Cf	2.6 y	5×10^{12} n s^{-1} g^{-1}	[(d)]

[(a)] Dose rate at 1 m from source, cf. eqn. (7.9).
[(b)] Variable source of $Pu^{16}O_2 + H_2$ ^{18}O; by transferring water through heating into the porous oxide, the reaction ^{18}O(α,n)^{21}Ne is initiated; by cooling the water evaporates and is returned to storage whereby the n output decreases to < 0.01 of maximum output.
[(c)] The half-life can be increased to 22 y by including the precursor nuclide ^{210}Pb.
[(d)] γ-dose rate increases with time due to formation of γ-emitting daughters.